SERVICE CIRCULAR

TATA MOTORS

SC/ 2025 /36	Model : LPT 1612g with 3.8 SGI TC CNG BS6 Phase-2 Truck	Group : 00	Mar 2025

All Dealers / TASSs',
We are pleased to inform introduction of LPT 1612g with 3.8 SGI TC CNG BS6 Phase-2 Truck.

Representative picture of the vehicle is given below -

We are enclosing the following details:

- Chassis type designation **(Annexure - 1)**
- Technical Specifications **(Annexure - 2)**
- Warranty and Free Service Schedule **(Annexure - 3)**
- Salient Features **(Annexure - 4)**
- BS6 Phase-2 Exhaust System and New Sensors **(Annexure - 5)**
- New Instrument Cluster **(Annexure - 6)**
- GDCU and FOTA Flashing **(Annexure - 7)**
- Cruise Control Function **(Annexure - 8)**
- Service and Maintenance Schedule **(Annexure - 9)**
- Recommended Oil and Lubricants **(Annexure - 10)**
- Special Tools List **(Annexure - 11)**
- EMS Diagnostic Software Details **(Annexure - 12)**

(*As per policy of Tata Motors to continuously improve their products, the company reserves the right to make changes of any nature on vehicles and aggregates without any obligation to incorporate them on previous vehicle)*

Annexure - 1
CHASSIS TYPE DESIGNATION:

Model	VC No.	Chassis barrel
1612g LPT DCR39CBC 125B6M6	29000139000R	563042
1612g LPT DCR39HSD 125B6M6	29000239000R	563043
1612g LPT DCR45CBC 125B6M6	29004045000R	563080
1612g LPT DCR45HSD 125B6M6	29004145000R	563081
1612g LPT DCR49CBC 125B6M6	29004049000R	563082
1612g LPT DCR49HSD 125B6M6	29004149000R	563083
1612g LPT DCR53CBC 125B6M6	29004053000R	563084
1612g LPT DCR53CBC 125B6M6	29004153000R	563085
1612g LPT DCR49CBC 125B6M6	29003149000R	563101
1612g LPT DCR49HSD 125B6M6	29003849000R	563102

Annexure - 2
TECHNICAL SPECIFICATIONS

Vehicle Model	1612g LPT DCR49CBC 125B6M6 CX	1612g LPT DCR49HSD 125B6M6 CX	1612g LPT DCR49CBC 125B6M6	1612g LPT DCR49HSD 125B6M6	1612g LPT DCR45CBC 125B6M6
VC No	**29003149000R**	**29003849000R**	**29004049000R**	**29004149000R**	**29004045000R**
Chassis Type	**563101**	**563102**	**563082**	**563083**	**563080**
ENGINE					
Engine Model	TATA 3.8 SGI TCIC BSVI				
Engine Type	4 Cylinder Inline, water cooled, turbo-intercooled				
Bore and Stroke	97 mm x 128 mm				
Engine Capacity (cc)	3783 cc				
Nos. of cylinders	4 Inline				
Compression ratio	11.5 ± 1 : 1				
Max Power	92 kW @2250 rpm as per MOST/CMVR TAP 115/116				
Max Torque	500 Nm at 1400-1600 rpm as per CMVR TAP 115/116				
Firing Order	1-3-4-2				
Turbo Charger	Yes				
HP Pump	Common rail with electronic injectors with automatic advance				
Air Filter	Dry type remote mounted				
Fuel Filter	Particulate Filter				
Oil Filter	Cartridge Filter				
Capacity Of Cooling System	18 liters				
Coolant	Water & Ethylene glycol, ratio 60:40, premixed				
Engine Fan Type	E-Viscous fan				
Exhaust System	Compatible with BS-VI & RDE Norms.				
FUEL SYSTEM					
Type	CNG Fuel System, CNG cylindrical Tank				
CNG Tank volume (Water Capacity)	550 Liters	550 Liters	440 Liters	440 Liters	440 Liters
CLUTCH					
Outside diameter of clutch	330 mm Multistage				
Engine Clutch Type	Single plate dry friction type				
Type of Actuation	Hydraulically actuated with pneumatic assistance.				
GEARBOX					
Gear box Model	GBS-550 (6.9) Synchromesh with PTOP, Cable shift				
No. of gears	6 Forward 1 Reverse				

Gear Ratios	1st- 6.9 , 2nd- 4.02, 3rd- 2.39 , 4th- 1.46 , 5th- 1.00, 6th – 0.84, Rev- 6.37.				
PROPELLER SHAFT					
Type	SPL055				
REAR AXLE					
Rear Axle Description	Single reduction hypoid gears, fully floating axle shafts.				
Rear Axle Type & Model	Banjo RA 108RR				
Rear Axle Ratio	6.43 (45/7)				
Rear Axle Make	TML				
FRONT AXLE					
Front Axle Description	Heavy duty Forged I beam, Reverse Elliot type				
Axle Make	TATA				
Front Axle Designed Capacity	5.6T				
ELECTRICAL					
Battery	100 Ah capacity				
Alternator Capacity	120 Amps, Multi groove				
Starter Motor	12V,2.2KW,TCO				
Head Lamp	2 No's with 1.8% Inclination with HLL				
Horn	1 Horn				
EMS	Westport				
WHEELS & TYRES					
Tyre Size & Tyre Type	9.0R20 Tube Tyre				
Front Tyre	9.0R20 -16 PR RIB				
Rear Tyre	9.0R20 -16 PR LUG				
No. Of Tyres	Front - 2,Rear-4,Spare-1				
Wheel Rims	7 x 20				
FRAME					
Frame Description	Ladder type heavy duty frame with riveted/bolted cross members Side members are of channel section				
	Depth - 220 mm (max), Flange Width - 60 mm, Thickness – 7 mm				
BRAKES					
Service Brakes	Fully duplicated, full air S-CAM brake system				
Front Brakes	Brake Liners - 410(Dia) x 180 (width)				
Rear Brakes	Brake Liners - 410(Dia) x 180 (width)				
Parking brake Description	Hand Operated, Spring actuated parking brake acting on rear wheels				
STEERING					
Type	Power Assisted hydraulic type				
Ratio	20.2:1 ZF, 20.4:1 Rane				
Steering Gear Box	Re-circulating Ball type				
SUSPENSION					
Front Suspension	Parabolic spring				
Rear Suspension	Leaf spring				
Shock Absorber	Hydraulic double acting telescopic type at front only				
Main Chassis Dimensions in mm					
Vehicle Overall Height (Unladen)	2520	3200	2520	3200	2520
Vehicle Overall Length	8860	8950	8860	8950	8320
Vehicle Overall Width	2270	2270	2270	2430	2270
Ground Clearance	~274mm				
Minimum Turning Circle Dia.	18900	18900	18900	18900	15600
Wheel Base	4920	4920	4920	4920	4530
PERFORMANCE					
Max. Geared speed	70 KMPH				
Max. Gradeability at rated GVW	27.00%				
WEIGHT in Kg					
Gross Vehicle Weight (GVW)	16371				
Kerb Weight (Unladen Weight)	4975	5935	4975	5935	4785
Max Payload	11336	10436	11336	10436	11586
Max. Permissible FAW	5671				
Max Permissible RAW	10700				

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Air Filter	Dry type remote mounted				
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Outside diameter of clutch	330 mm Multistage				
Engine Clutch Type	Single plate dry friction type				
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Axle Make	TATA				
Front Axle Designed Capacity	5.6T				
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Alternator Capacity	120 Amps, Multi groove				
Starter Motor	12V,2.2KW,TCO				
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Tyre Size & Tyre Type	9 R20, Tube Tyre				
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Vehicle Overall Length	8320	7360	7400	9530	7400
Vehicle Overall Width	2430	2270	2430	2270	2430
Ground Clearance	~274mm				
Minimum Turning Circle Dia.	15600	13800	13800	18100	13800
Wheel Base	4530	3920	3920	5300	5300
PERFORMANCE					
Max. Geared speed	70 KMPH				
Max. Gradeabilty at rated GVW	27.00%				
WEIGHT in Kg					
Gross Vehicle Weight (GVW)	16371				
Kerb Weight (Unladen Weight)	5620	4845	5635	4780	4870
Max Payload	10751	11526	10736	11591	11691
Max. Permissible FAW	5671				
Max Permissible RAW	10700				

Annexure – 3
WARRANTY AND FREE SERVICE SCHEDULE

3.1 Warranty:

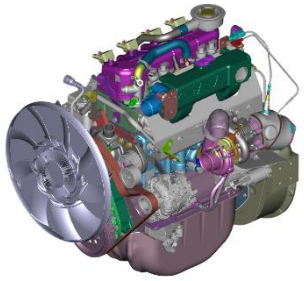
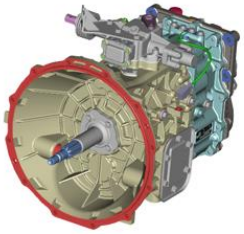
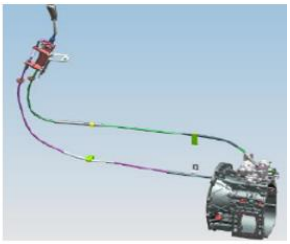
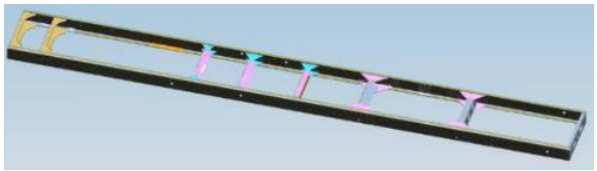
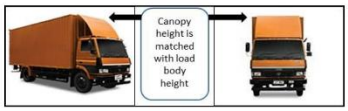
Model	Vehicle Warranty	Engine Warranty
LPT 1612g with 3.8 SGI TC CNG BS6 Phase-2 Truck	36 months from the date of sale OR 3,00,000 Km whichever expires earlier	Same as Vehicle Warranty

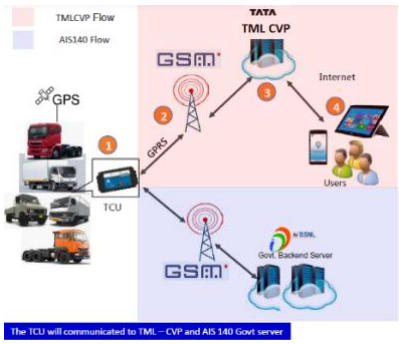
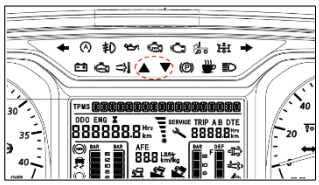
3.2 Free Service Schedule:

Below free services are applicable at the interval given below:

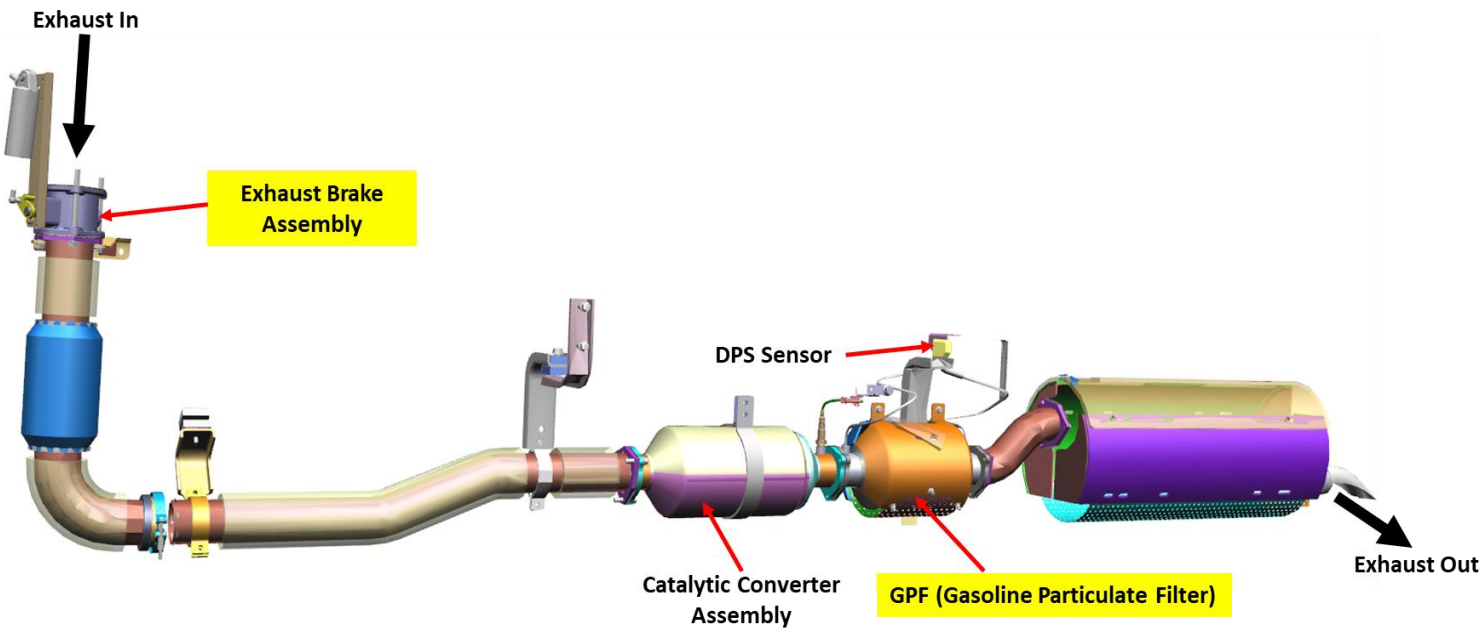
Sl. No	Service	Kms covered (± 2000)	Days (± 60)
1	**Free PDI**	**At the time of delivery of vehicle**	
2	**Free Wheel Alignment**	**3,000 to 5,000 Km**	**NA**
3	**First Free Service**	**20,000 Km**	**< 365 Days (1 Year)**
4	**Second Free Service**	**40,000 Km**	**< 730 Days (2 Year)**
5	**Third Free Service**	**60,000 Km**	**< 1095 Days (3 Year)**

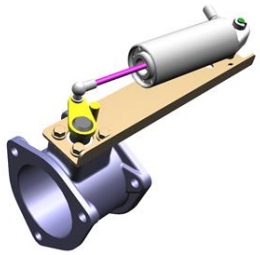
Annexure – 4
SALIENT FEATURES

SN	Descriptions	Photo
1	**3.8 SGI TC BS6 Engine** ➢ 3.8 SGI TC BS6 Engine with max engine output 92 kW @ 2250 rpm	
2	**Gear Box-G550 - 6 Speed** ➢ Cable shift gearbox with reduced shifting efforts	 
3	**Modular Chassis Frame** ➢ Straight ladder type frame	
4	**Cabin** ➢ Walk Through Cabin ➢ Driver Suspended Seats ➢ Tilt & tel steering column	
5	**Air Deflector Mounting Provision** ➢ Total 8 , M8 tapped holes ➢ Better FE	
6	**GDCU – Gateway Domain Controller Unit** ➢ Prevents unauthorized access to vehicle CAN network ➢ Performs FOTA function. Flashing of vehicle various ECUs remotely.	

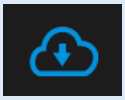
7	**Telematics** ➢ Fleet Management Solution ➢ Track and Trace ➢ Driving Behavior ➢ Vehicle Performance	
8	**Cruise control Function** (If applicable) ➢ Helps driver to maintain a suitable vehicle speed without keeping foot on the accelerator pedal. **Note**: At least once **Brake** should have been pressed after starting the vehicle to activate Cruise Control.	
9	**FOTA** (Firmware-update Over The Air)	
10	**GSA** (Gear Shift Advisory) (If Applicable) ➢ Gear Shift Advisor or indicator helps notify the driver to display the current gear during manual gear shifting and when it is appropriate to engage the next gear for optimum fuel economy.	

Annexure – 5
BS6 Phase-2 Exhaust System and New Sensor



New Sensor / Actuator		
Sr. no.	**Description**	**Image**
1	**GPF (Gasoline Particulate Filter)** ➢ Gasoline particulate filters (GPF) are based on the proven wall-flow technology of diesel particulate filters. While the purpose of DPF is to mitigate particulate emissions from diesel engines, GPFs were developed to reduce fine particulate emissions from gasoline vehicles.	
2	**Exhaust Brake** ➢ Exhaust brakes can prevent a vehicle from going downhill too fast. ➢ If a driver transports a loaded vehicle and he needs to go downhill, the exhaust brake can prevent the vehicle from going too fast and it reduces the likelihood of an accident.	

Annexure – 6
New Instrument Cluster

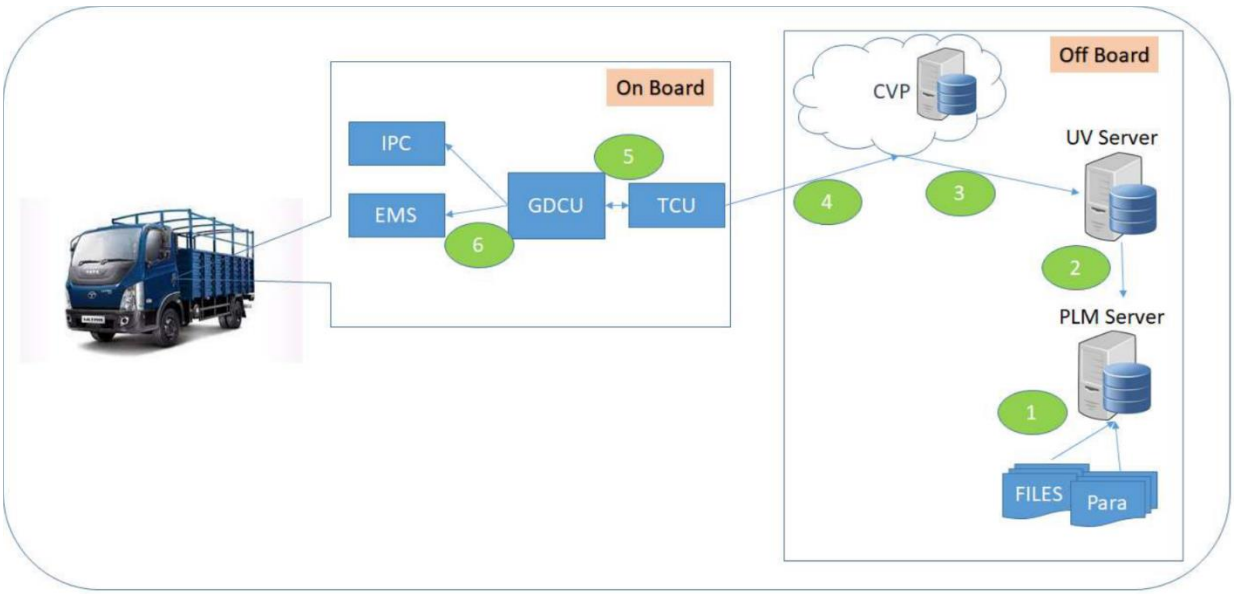
FOTA (Firmware-update Over The Air)		It will illuminate for approx. 3 sec when ignition is switched 'ON'. This Indicator Turns "ON" when Firmware Over The Air Update (FOTA) is Available & process initiation required. This Indicator will start "blinking" once the FOTA activity is initiated.
Cruise Control Function		It will illuminate when cruise control function in 'ON'.

Annexure – 7

GDCU and FOTA Flashing

- GDCU stands for Gateway Domain Controller Unit. It acts as gateway between various CAN network ECUs/controllers
- It prevents unauthorized access to vehicle's CAN network.
- GDCU Enables FOTA (Firmware over the Air) function. Any vehicle ECU can be flashed remotely without visit to any workshop.

Architecture of GDCU and FOTA function



- Updated Calibration firmware released on PLM server, gets transferred to UV server.
- From UV server, Cloud CVP (Connected Vehicle Platform) transfers firmware to particular chassis TCU (Telematics Control Unit). Updated firmware gets stored in memory of GDCU.
- Driver will get intimation on instrument cluster by FOTA Tell Tale lamp about new software available.
- Pre-Conditions for FOTA flashing are to be satisfied to initiate process.
  - Hand Brake Engaged
  - Engine RPM = 0 , Vehicle speed = 0
  - IGN – ON
  - Battery Voltage – 12V
- Once pre-conditions are satisfied, drive has to press Hazard switch 4 times (ON-OFF-ON-OFF) within 10 sec to start software update.
- After driver authentication, GDCU will flash particular ECU.

- Cluster will display following ID which denote which ECU being updated

Vehicle ECU	ID on cluster
EMS (Bosch)	06
EMS (Delph)	01
GDCU	20
TCU	25
Cluster	08/09
VECU	13

- Flashing status and progress will be displayed on screen.
  - **FOTA STARTED** text message shall appear only once for 4 Secs
  - **TIME : XXXMINS** text message shall appear for 4 Secs
  - Cluster screen will go blank and ECU ID for which FOTA is initiated will be displayed.
  - Cluster will display **FOTA-XX-XXX%** percentage completion of software update
  - During FOTA in progress, FOTA tell-tale will keep on blinking

- After successful flashing **FOTA SUCCESS** message will display on screen. Driver can drive the vehicle. If software update is unsuccessful then **FOTA FAIL - ECU NOT OK** or **FOTA FAIL - ECU OK** message will display on screen.
- If FOTA fails then previous version flashing will be initiated with message on screen as **FOTA ROLLBACK** along with status and progress. Either **ROLLBACK SUCCESS** or **ROLLBACK FAIL** message will appear accordingly. If ROLLBACK FAIL message appear on display; visit to nearest Tata motors Authorized Service Station.

- After **FOTA COMPLETE** message, tell-tale FOTA lamp will disappear from cluster.

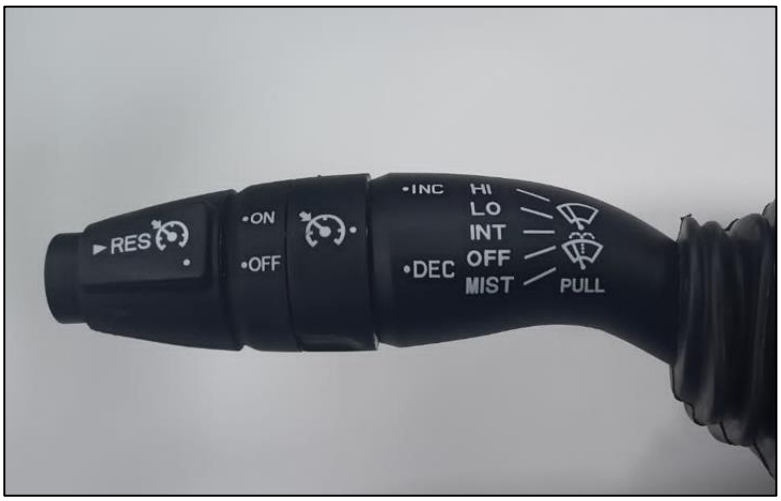
Do's	Don'ts
Regularly check for this tell-tale indication	Do not crank vehicle during software update in progress.
Educate the drivers to keep a watch for the tell-tale and guide them to authorize through hazard switch	Do not turn off ignition during software update in progress
Follow pre-condition requirement	Do not cut-off battery / Do not turn off battery switch during software update
Perform the software update when vehicle is parked in safe location and safe condition	Do not perform software update if battery voltage level is low
	DO not leave vehicle unallocated while software update is in -progress

*IPC: Instrument Panel & Cluster
*EMS: Engine Management System

Sample Display Message	Display image
TIME : XXXMINS	
ROLLBACK SUCCESS	
FOTA COMPLETE	

Annexure – 8

Cruise Control Function (If Applicable)



- Cruise Control function helps driver to maintain a suitable vehicle speed without keeping foot on the accelerator pedal.
- Cruise control function can be set **ON** or **OFF** by rotating left knob
- When cruise control is ON, Tell-tale lamp will glow in instrument cluster.
- Speed can be increased or decreased by 1 Km/hr. by rotating knob in front of **INC (Increase)** and **DEC (Decrease)**
- To cancel cruise control temporarily, press either brake pedal or clutch pedal.
- Cruise control can be resumed by pressing **RES (Resume)** button.

Preconditions to start cruise control function

- Desired constant vehicle speed is above 40km/ph.
- Vehicle is in 4th or 5th gear.

Temporary Disengagement of cruise control

- Brake pedal or Clutch pedal is pressed
- Speed falls below 30 km/h, when travelling uphill
- Gear is changed

Automatic Disengagement of cruise control

- Vehicle speed is below 30 km/h.
- Low engine speed (< 1000 rpm)
- Vehicle speed sensor is disconnected or tampered or non-functional.

Annexure – 9

SERVICE AND MAINTENANCE SCHEDULE

9.1 Service Operations to be performed by Customer –

SN	OPERATION	FREQUENCY
1	Check engine oil and coolant level. Top up, if necessary.	Daily
2	Carry out visual inspection of vehicle for leakages, traces of oil / fluid on ground, if any.	Daily
3	Check for switching sound of gas shut off solenoid valve after inserting ignition key.	Daily
4	Check whether CNG Fuel gauge in instrument cluster is indicating correctly, compared to high pressure gauge.	Daily
5	Check air filter service indicator for the red band. Contact nearest TATA Motors Authorised Workshop if red band appears. **Note - Please do not clean air filter element.**	Daily
6	Check for proper functioning of switches, gauges, warning lamps on instrument cluster, buzzers, all lights, other electrical controls.	Daily
7	Check for proper functioning of service brake, parking brake, LCRV and exhaust brake.	Daily
8	**Drive Belts** - Inspect the belts weekly. Check the belt for intersecting cracks. Attend, if required.	Weekly
9	**Air Tanks and Reservoirs (For Air Brake only)** - Open the drain cock on the air tank to drain any moisture accumulated in the air system. If oil is present, the air compressor system must be checked at TATA Motors authorised workshop.	Weekly
10	Check catalytic converter housing exterior for burn marks / damage.	Weekly
11	To check proper functioning of cylinder valves, close all the cylinder valves and start the engine. If engine does not stop by itself, get valves checked / replaced. Check that the pressure gauge shows full reading now	Weekly
12	Check tyre pressure including spare wheel, inflate tyres to specified pressure if necessary.	Weekly
13	Check clutch, power steering oil level. Top up, if necessary.	Monthly
14	Visual inspection of battery terminal/ post condition, electrolyte level, battery cable connections, battery mountings, If there are any abnormalities observed during visual inspection, request to visit at TATA Motors authorized workshop.	Monthly / 5000 Kms.
15	Check tyres for any abnormal tyre wear. If tyre wear noticed, wheel alignment to be checked immediately.	Monthly / 5000 Kms.
16	Lubricate all chassis grease nipple points, if required	Monthly / 5000 Kms.
17	Rotate the tyres as per recommendation. In case any abnormal tyre wear observed, rotation to be carried out immediately for better tyre life.	Every 20000 Kms.
18	Check the wheel alignment.	Every 20000 Kms.
19	Check head lamp focusing. Adjust if necessary.	As required
20	Check windshield washer water level. Top up, if necessary	As required
21	Wash the vehicle	As required

9.2 Service Operations to be performed at Workshop:

SERVICE OPERATIONS TO BE PERFORMED AT WORKSHOP						
D=Daily, W=Weekly, M=Monthly, P=PDI, *=whichever is earlier						
SN	**OPERATION**	**FREQUENCY**	**Checks by Customer**			**PDI**
			D	**W**	**M**	
1	**ENGINE**					
1.1	Drain off Engine oil while hot. Change engine oil and oil filter. Clean drain plug & engine breather. Check engine oil pressure.	20,000 Kms. OR 1 Year*				
1.2	Check condition of drive belts for water pump, alternator, fan, steering visually. Replace, if necessary.	20,000 Kms.		**W**		**P**
1.3	Check all drive belts for free rotation / damage: (a) Water pump (b) Belt Tensioner pulley (c) Belt Idler pulley (d) Fan pulley (e) Alternator pulley. Change if necessary.	20,000 Kms.				
1.4	Belt drive parts have to be compulsory changed.	80,000 Kms. OR 2 Years*				
1.5	Check if red band appears on air filter Service indicator or 'Air filter clogging' indicator on instrument cluster is ON. Replace if necessary	20,000 Kms.	**D**			**P**
1.6	If red band appears on service indicator, replace primary filter element. Replace secondary (Safety filter) during every 3rd replacement of primary filter. **(Note - Please do not clean air filter element with air)**	20,000 Kms.				
1.7	Check proper sealing of air intake and exhaust system by checking hose, gaskets & pipe condition / proper tightness of clamps. Replace any defective item, if required.	20,000 Kms.				
1.8	Check / replace if required pipes, clamps & hoses of engine lubrication / fuel / coolant / windshield washing system.	20,000 Kms.				
1.9	Replace rubber hoses of engine lubrication / coolant. Take the vehicle to workshop if any leakage is noticed.	80,000 Kms. OR 2 Years*				
1.10	Check cylinder head valve clearance (Tappet setting) & adjust, if necessary.	20,000 Kms.				
1.11	Check for external clogging of radiator, intercooler, and condenser (If fitted), clean if necessary with compressed air & water.	20,000 Kms.				
1.12	Replace Spark Plugs.	20,000 Kms.				
1.13	**Rubber boot along with Spring of Ignition coil:** Clean rubber boot and refit. Replace it if found cracked/damaged	20,000 Kms				
1.14	Replace rubber boot of ignition coil along with Spring.	40,000 Kms				
1.15	Check coolant concentration and top up coolant, if required.	20,000 Kms.				
1.16	**Drain cooling system** - Reverse flush. Check thermostat for proper operation. Refill system with fresh coolant. Use mixture of clean water & recommended anti-freeze agent in specified ratio.	300,000 Kms. OR 2 Years*				
1.17	Check thermostat working & fan operation.	20,000 Kms.				
1.18	Check for radiator mounting rubber bush. Replace, if required. Check & tighten radiator mounting.	20,000 Kms.				

1.19	Check and tighten engine mounting rubber, if required.	20,000 Kms.				
1.20	Check mounting of engine speed sensor, boost pressure sensor, air intake temperature sensor, coolant temperature sensor, vehicle speed sensor.	20,000 Kms.				
1.21	Check cylinder head cover gasket condition for leakage and change if necessary.	20,000 Kms.				
1.22	Check for ECU errors with diagnostic system based on error indication in cluster, if any.	20,000 Kms.				
1.23	Check catalytic converter housing exterior for burn marks / damage.	20,000 Kms.		**W**		**P**
2	**FUEL SYSTEM**					
2.1	Check for switching sound of gas shut off solenoid valve after inserting ignition key	20,000 Kms.	**D**			**P**
2.2	Check whether CNG Fuel gauge in instrument cluster is indicating correctly, compared to high pressure gauge.	20,000 Kms.	**D**			**P**
2.3	To check proper functioning of cylinder valves, close all the cylinder valves and start the engine. If engine does not stop by itself, get valves checked / replaced. Check that the pressure gauge shows full reading now	20,000 Kms.		**W**		**P**
2.4	Check for corrosion on all CNG cylinders. Rectify, if corroded.	20,000 Kms.				**P**
2.5	Check lambda sensor output (milli volt variation) its functionality.	20,000 Kms.				
2.6	Replace CNG low pressure hoses.	80,000 Kms. OR 2 Years*				
2.7	Replace high pressure gas filter cartridge and body O-ring.	60,000 Kms.				
2.8	Clean HP gas filter and replace O-ring. **Filter cleaning procedure:** Close all cylinder valves and start the engine. Engine should stop on its own confirming that there is no gas in the system. Open high pressure filter cap and separate it from HP Filter body. Dismantle. Clean with air and refit.	20,000 Kms.				
2.9	Replace the O-rings of CNG receptacle	300,000 Kms. OR 5 Years*				
2.1	Clean High Pressure Solenoid Valve inlet filter.	60,000 Kms.				
2.11	Replace High Pressure Solenoid Valve inlet filter.	100,000 Kms.				
2.12	Recertification of cylinders as per gas cylinder rules to be carried out by competent authorities.	Every 3 Years				
3	**CLUTCH**					
3.1	Check for free play movement of clutch pedal. Adjust if necessary.	20,000 Kms.				
3.2	**Gear Shifting** - Drive the vehicle by shifting all gears to ensure clutch functioning (engagement, disengagement and torque transmission) and replace appropriate components if found defective.	20,000 Kms.				
3.3	Replace clutch fluid & carry out bleeding of system. Bleeding of clutch actuation system, In case of any complaints please check / replace affected parts.	80,000 Kms.				
3.4	Overhaul clutch master, slave cylinder / clutch booster.	80,000 Kms.				
4	**GEAR BOX**					
4.1	Change oil in Gear Box. Drain while hot. Clean drain plug & breather.	120,000 Kms.				
5	**FRONT AXLE**					

5.1	Remove front hub caps, fill 3/4th full with wheel bearing grease & refit. **(For Non THU Bearing if applicable)**	20,000 Kms				
5.2	Grease with grease gun: (a) Kingpins, (b) Tie rod ends, (c) Drag link ends	20,000 Kms				
5.3	Remove front wheel hubs. Dismantle and clean bearing & other components. Replace damaged / worn out parts. Replace with fresh wheel bearing grease and refit. Adjust wheel hub bearing play. **(For Non THU Bearing if applicable)**	1,60,000 Kms				
6	**REAR AXLE**					
6.1	Change oil in rear axle. Drain while hot. Clean drain plug and breather.	3,00,000 Kms				
6.2	Check and adjust hub play, top up the grease (If Required) **(For Non THU Bearing if applicable)**	20,000 Kms				
6.3	Remove rear wheel hubs. Dismantle and clean bearings and other components. Replace damaged/ Worn-out parts. Repack with fresh wheel bearing grease and refit. Adjust wheel hub bearing play. **(For Non THU Bearing if applicable)**	160,000 Kms.				
6.4	Adjust crown wheel thrust pads	20,000 Kms				
For THU Bearing - Front and Rear Axle Wherever applicable						
1	Inspection of front / rear hubs (wheel) 1. Check for any abnormal noise. 2. Check bearing hub play (Spec:-Hub play less than 120 micron). 3. Check grease leakage for front and oil leakage for rear by opening the HUB CAP. 4. Check for any water ingress by removing the Hub Cap. **(DO NOT DISTURB THE UNITIZED BEARING AND INTEGRATED SEAL)**	20,000 Kms				
2	Hub greasing is not required on THU bearing fitted vehicle.	NA				
7	**PROPELLER SHAFT**					
7.1	Check Propeller shaft splines for excessive wear and noise. Replace, if necessary.	20,000 Kms.				
8	**BRAKES (For both Hydraulic and Air Brake System)**					
8.1	Check for proper functioning of service brake, parking brake, LCRV and exhaust brake (if applicable). Adjust, if necessary.	20,000 Kms.	**D**			**P**
8.2	During tyre rotation (with the wheels removed), Check for front & rear brakes uneven wear of brake linings and grease spilling on linings or past hub seals. Check all springs of brake shoe for no elongation. Attend, if necessary.	20,000 Kms.				
	FOR HYDRAULIC BRAKES					
8.3	Change brake fluid & bleed the system.	40,000 Kms.				
8.4	Check functioning of Vacuum Booster, Tandem Master Cylinder & Wheel Cylinder. Overhaul, if necessary.	80,000 Kms. OR 2 Years*				
	FOR AIR BRAKES					
8.5	Drain off air from all tanks. If condensed water is found, replace air drier desiccant cartridge.	20,000 Kms.		**W**		**P**
8.6	Check for oil droplets / excessive oil accumulation (slight traces of oil are acceptable) at air drier exhaust port. If yes, then carry out following activities: (a) Check / replace compressor piston rings, (b) clean	20,000 Kms.				

	compressor head (c) Check / replace compressor outlet pipe in case of carbon formation, (d) clean oil separator.					
8.7	Check for air dryer outlet pressure difference between cut-off and cut-in, which should be within 0.6 bar to 1.3 bar.	20,000 Kms.		**W**		**P**
8.8	Inspect torque plate, cam shaft, shoe seemly, dust shield, chamber bracket assembly, roller and fulcrum pin. Replace if required.	40,000 Kms.				
8.9	Conduct brake system routine **checks I** (Check for leakage, switches and gauges, brake pressure fill up time).	20,000 Kms.	**D**			**P**
8.10	Conduct brake system routine **checks II** (Check opening / closing pressure of SPV, check valves for leakage, damage etc.)	200,000 Kms. OR 2 Years*				
8.11	Dismantle and overhaul pneumatic aggregates of brake system. Clean / inspect & replace parts if necessary.	200,000 Kms. OR 2 Years*				
8.12	Replace air drier desiccant cartridge and filter of air dryer / DDU.	200,000 Kms. OR 2 Years*				
9	**SUSPENSION**					
9.1	Check front & rear spring bushes, anti-roll bar bush, Cab spring eye bushes, replace if required.	40,000 Kms				
9.2	Check suspension. Attend, if required - Leaf springs for damage / sagging / bush wear out / U-bolt suspension arm & steering knuckle supports & rear leaf spring arm. Suspension bump stop.	40,000 Kms				
9.3	Dismantle front & rear spring & cab mounting spring. Clean and inspect leaves. Apply graphite grease on leaves & re-assemble.	Every 1,20,000 Kms.				
9.4	Check shock absorbers for leak / damage. Check shock absorber bush, replace if required.	40,000 Kms		**W**		**P**
9.5	Check antiroll bar for damage / bush wear out.	40,000 Kms				
10	**STEERING (POWER / MECHANICAL)**					
10.1	Check steering pipelines connections for leakages and rectify if any.	20,000 Kms.				
10.2	Check power steering hydraulic limiter valve setting and peak pressure.	40,000 Kms.				
10.3	Check steering wheel play & check for proper condition of ball joints & rubber boots of tie rod / drag link for looseness, damage, excess wear, dislodging etc. Replace if necessary.	40,000 Kms.				
10.4	Drain off hydraulic oil of power steering / mechanical steering. Replace filter cartridge. Fill in fresh oil.	160,000 Kms. OR 2 Years*				
11	**WHEELS AND TYRES**					
11.1	Check tyre pressure; inflate tyres to specified pressure, if necessary.	Daily				
11.2	Rotate the tyres as per recommendation.	20,000 Kms				
11.3	Check for wheel nut tightness, for specified torque.	During every wheel change & 100 Km. after wheel change				
11.4	Check wheel alignment. (**Note** - Wheel alignment to be done even after changing related suspension & steering linkage parts)	20,000 Kms OR if any abnormal tyre wear noticed*				
12	**ELECTRICALS**					
12.1	Check proper functioning of switches, gauges, lights, blinkers, warning lamps in instrument cluster.	20,000 Kms.	**D**			**P**

12.2	Check battery mountings, Clean battery posts and terminals, tighten terminals & smear petroleum jelly. Check battery cells with cell tester and specific gravity of electrolyte with hydrometer. Check electrolyte level in battery and top up if necessary.	20,000 Kms.				**P**
12.3	Check horn mounting and functioning. Adjustment to be done if required.	20,000 Kms.				**P**
12.4	Check head lamp focusing. Adjust if necessary.	20,000 Kms.	**D**			**P**
12.5	**Check all electrical system health:** Ensure usage of genuine fuses with correct rating, All ECU, DC-DC converter, telematics device, earthling connections, proper routing / clamping of wiring, intactness of grommets.	20,000 Kms.				
12.6	Check sensors & connectors for cleanliness & secure fitment.	After every wash				
13	**CAB & LOAD BODY**					
13.1	Check for any paint peeling, rusting & damage to body panels & structure all around & underneath.	20,000 Kms.				**P**
13.2	Check for proper functioning of doors, seat, and sun visors. Check body for unusual noise.	20,000 Kms.				**P**
14	**GENERAL**					
14.1	Check & top up levels of: 1. Engine oil 2. Engine coolant 3. Windshield washer tank (As required)	20,000 Kms.	**D**			**P**
14.2	Check & top up oil levels of: 1. Gear Box oil 2. Rear Axle oil 3. Mechanical Steering oil	20,000 Kms.				**P**
14.3	Check & top up oil levels of: 1. Clutch oil 2. Brake oil (If Applicable) 3. Power steering oil (If Applicable)	20,000 Kms.			**M**	**P**
14.4	Check for / Rectify the leakages of 1. Oil (at Engine / Clutch / Gear Box / air conditioning (If fitted) / shock absorbers / Rear axle/ power steering) 2. CNG at tank, pipelines & engine 3. Engine coolant 4. Engine exhaust 5. Pneumatic (brake system)	20,000 Kms.	**D**			**P**
14.5	Check and tighten all fasteners if necessary as per the check list #	20,000 Kms.				
14.6	Wash the vehicle.	As Required				
14.7	Grease as per the list ##	20,000 Kms.			**M**	**P**
14.8	Conduct road test for vehicle handling and functioning of aggregates.	20,000 Kms.				
14.5	**# Checking of fasteners required Periodical Checking and Tightening**					
a	**Engine peripherals** : Engine mounting & accessories mountings, all clamps, CNG tank brackets, air filter mounting, radiator mounting, thermostat, exhaust muffler, exhaust pipe mounting hangers & brackets, air duct hose connections.	20,000 Kms.				
b	**CNG Gas circuit** : Mountings of 3 way valve / CNG pressure transducer / receptacle / HP filter / LP filter / HP regulator / HP solenoid valve / Gas cylinder	20,000 Kms.				

c	**Driveline**: Clutch housing, clutch booster, mounting of clutch master cylinder and slave cylinder, gear box mounting, gear shift mechanism, propeller shaft coupling flange, center bearing bracket, rear axle carrier housing mounting studs.	20,000 Kms.				
d	**Steering**: Steering gear box bracket mounting bolts, pitman arm, drag link, tie rod, relay arm, steering column.	20,000 Kms.				
e	**Suspension** : Anti roll bar mounting, shock absorber mounting, leaf springs 'U' bolts, nuts of spring wedge bolts, spring hanger bracket & slide support bracket	20,000 Kms.				
f	**Brakes** : Wheel cylinders, Brake chambers, spring actuators & other brake valves mountings, hydraulic / pneumatic lines, anchor plate, air tanks mounting, airline clamps, exhaust brake valve mounting bolts, relay valve mounting, solenoid valve mounting, modulator valve mounting.	20,000 Kms.				
g	**Electricals**: Ignition coil connectors, starter motor, and alternator, battery mounting bolts, wiper motor, switches & gauges, head lamp, tail lamp, blinker lamp.	20,000 Kms.				
h	**Wheels & Tyres**: wheel mounting nuts, mounting of spare wheel carrier.	20,000 Kms.				
i	**Cab** - Cab mountings, mounting of cab lock and cab springs, Seats, RVM, door hinges, door locks, door striker plates, body mountings.	20,000 Kms.				
j	**General**: Apart from above please check visually any other fasteners for loosening.	20,000 Kms.				
14.7	**## List of items required periodic Greasing**					
a	**Clutch** : Clutch release bearing, clutch cross shaft, clutch pedal bush	20,000 Kms.			**M**	**P**
b	**Front Axle** : King pin	20,000 Kms.			**M**	**P**
c	**Prop shaft** : UJ cross joints, sliding yoke (also check the condition of rubber boot), center bearing	20,000 Kms.			**M**	**P**
d	**Steering** : Steering tie rod ends, ball joints, drag links, center link, idler arm, spindle / sleeves, column	20,000 Kms.			**M**	**P**
e	**Brake:** Front and rear Brake S-camshaft bushes. (For air brake only)	20,000 Kms.			**M**	**P**
f	**Lubricate with oil can -** Pinion bushing of starter motor and wiper & motor linkages. Door hinges, outer door handle, door latches, dove tails & striker plates, bonnet stay rod, pivot pins & luggage box door	20,000 Kms.			**M**	**P**

Annexure – 10

RECOMMENDED OIL AND LUBRICANTS

SN	AGGREGATES	QUANTITY	FLUID TYPE / GRADE	RECOMMENDDED BRANDS
1	Engine oil (including oil filter)	**A - During scheduled services** - 14.5 Ltrs (Including oil filter). **B - For Green / New Engine Assy** (Including oil filter) -16 Ltrs. **Ensure oil topped up to max level**	TMGO Gas Engine Oil 15W40 LONG LIFE	**TMGOGEO15W40LL***
2	Coolant	18 Ltrs.	Non-Amino Base Coolant Concentrate meeting SS 7721 (Water & Ethylene glycol, ratio 60:40, premixed)	**TATA MOTORS** - Coolant Type 2, Ultra Long Life, SS7721 Part number - 55120860665
3	Gear Box (GB-550)	5.7 Ltr	SAE 80W90 Long drain	TATA MOTORS - TATA Motors Genuine Gear Oil 80W90 LL
4	Rear Axle (RA108 RR)	8.6 Ltrs	ProFE SAE 80W90 LL with SS6592	TATA MOTORS - TATA Motors Genuine ProFE LL 80W90 Axle Oil
5	Power Steering Oil	3 Ltrs.	Dexron II D	**TATA MOTORS** - TATA Genuine Power Steering Oil
6	Clutch Fluid	650 ml	DOT 4	**TATA MOTORS -** TATA Genuine Brake Fluid (DOT 4)
7	Wheel bearing grease **(For Non THU Bearing if applicable)**	Front Hub – 450gms/hub Rear Hub - 650 gms/hub	Synthetic Grease	**TATA MOTORS** - TATA Genuine Synthetic Grease
8	All Grease Nipples	As required	AP2 Grease	**TATA MOTORS** - TATA Genuine AP2 Grease

Notes:- For engine oil quantity related change details, Please refer below released service circular.

1) **SC / 2024 / 140** - Introduction of 14 Ltrs engine oil sump (Higher engine oil capacity sump) and amendment in engine oil quantity in LPT CNG PH2 trucks.

Annexure - 11
SPECIAL TOOL LIST

Note – For existing special tools, refer Workshop manual.

Newly added tools for BS6 Phase-2 vehicles

New Additional Special Tool					
Sr.No	**Aggregate**	**Tool Part No**	**Tool Description**	**Tool Image**	**Tool Use**
1	CNG Tank	5545 5890 4702	CNG Valve R&R tool		

Annexure - 12
EMS DIAGNOSTIC SOFTWARE DETAILS

All Workshops to ensure Garuda Hardware Kit available & Diagnostic Software is installed on Laptop to do diagnostic of all vehicle.

PART NAME	PART NUMBER	PHOTO
Laptop based Software TDS - ICV, HCV, I&LCV, SCV & PICK UPS Updated Version	210323999949	
TFS – ICV, HCV, I&LCV, SCV & PICK UPS Updated Version	210323859941	
Hardware Kit GES -Garuda II-USB_ ENABLED_FULL_SET Kit : (Garuda Hardware + USB Cable + OBDII DLC Cable)	210324769907	
Hardware Child Parts GES - Garuda II – USB Cable	210324769908	
Hardware Child Parts GES – Garuda II – OBD II – Cable (DLC CABLE)	210324769909	